Sentiment Analysis of Restaurant Reviews Case Study Rubric

Why am I doing this? This case study provides hands-on experience with a real-world classification problem that impacts both consumers and businesses in the restaurant industry. Through this project, you will learn how to work with imbalanced datasets, preprocess textual data, and train and evaluate machine learning models using R. By building a sentiment classification pipeline, you will develop practical skills in natural language processing (NLP), data preprocessing, and predictive modeling that directly translate to real-world industry applications and future data science projects.

What am I going to do? In this case study, you will build machine learning models that classify Yelp restaurant reviews by sentiment and identify which customer words and phrases are most strongly associated with positive or negative reviews. You will then evaluate and compare both models to determine which provides the best overall performance. Your deliverables will include:

- GitHub Repository containing code and necessary data
- A final report PDF summarizing the methodology, findings, insights, limitations, and future recommendations

Tips for success:

- Start with an EDA. Understanding your data is crucial before building your model.
- Start simple. Build a baseline model first, then experiment with preprocessing methods, features, and model improvements.
- Focus on the process, not just accuracy. Thoughtful data cleaning, feature engineering, and evaluation are just as important as model performance.
- Ask questions and share ideas. Discussing your approach with instructors or classmates can help strengthen your project and improve your understanding.
- Don't be afraid to experiment. Some approaches will work better than others, and that's a normal part of the machine learning workflow.

How will I know I have Succeeded? You will meet expectations on the Sentiment Analysis of Restaurant Reviews Case Study when you follow the criteria in the rubric below.

<table>
<tr><th>Spec Category</th><th>Spec Details</th></tr>
<tr><td>Code</td><td>Your code should include:

Data exploration and visualizations

Load and examine the Yelp Restaurant Reviews dataset
Analyze and visualize star rating distribution, review length, and word frequencies
Identify potential class imbalance in the dataset

Feature Engineering

Perform text preprocessing (e.g., lowercase, remove punctuation)
Perform TD-IDF feature engineering (e.g, tokenization, vectorizaiton)

Model Development
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Implement at least two classification models (e.g., Logistic Regression, Random Forest, Naive Bayes)
Use a stratified 80/20 train-test split
Address class imbalance if present (e.g., class weights, resampling)
Perform hyperparameter tuning for model optimization
Achieve at least 75% accuracy on at least one model

Evaluation and Analysis

Generate confusion matrices for all models
Compare accuracy, precision, recall, and F1-score
Plot feature importance for the top 20 features
Compare training vs test performance for each model
Evaluate ROC curves and ROC-AUC scores

Code should be submitted as a well-organized R script with clearly labeled sections. Code comments should explain key steps and decisions, and the script should run end-to-end without errors.
</td></tr>
<tr><td>Formatting</td><td>All materials should be submitted via Canvas using a GitHub repository. Your repository should include:

README.md — Brief summary of the case study and contents of the repository
LICENSE.md
Code files (R scripts for EDA, modeling, and analysis)
Dataset used for analysis
Outputs / Visualizations folder — Include tables, plots, and graphs
REFERENCES.md — All references formatted in IEEE style
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<tr><td>Final Report</td><td>The final report should be 2–3 pages (PDF format) and include:

Summary of approach and key findings
Explanation of feature engineering choices
Comparison of models, including strengths and weaknesses
Justification of final model selection
Discussion of limitations and future improvements
Key visualizations supporting conclusions
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<tr><td>References</td><td>All references should be listed in REFERENCES.md using IEEE citation style</td></tr>
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Helpful Resources:

[1] Grolemund, *A Installing R and RStudio | Hands-On Programming with R*. Accessed: May 04, 2026. [Online]. Available: https://rstudio-education.github.io/hopr/

[2] J. S. and D. Robinson, *2 Sentiment analysis with tidy data | Text Mining with R*. Accessed: May 04, 2026. [Online]. Available: https://www.tidytextmining.com/sentiment.html